Configuring Windows Autopilot for Device Enrolment

A step-by-step guide to streamline and standardize device enrolment via Microsoft Intune, ensuring secure and consistent deployment.

Tool: Microsoft Intune Portal

Author: Olu Adelokiki

Role: Lead EUC Engineer, UserCompute LTD

Website: www.usercompute.com

Contents

Step 1 Create a Device Group

Step 2 Configure Intune Policies and Apps

Step 3 Configure Enrolment Status Page

Step 4 Create Deployment Profile

Hardware ID Registration

STEP 1 Create a Device Group

My preference is to use a **Device Dynamic Group** to target all Autopilot devices in the organization. For this guide, we will call the group **"Sec - All Laptop/Desktops"**.

Use the following dynamic membership rule syntax:

```
(device.devicePhysicalIDs -any (_ -contains "[ZTDId]"))
```

This rule automatically captures all devices registered with a Zero-Touch Deployment ID (ZTDId), which is the identifier assigned when a device's hardware hash is uploaded to the Autopilot service.

STEP 2 Configure Intune Policies and Apps

Configure Device Policies

Set up device policies such as security baselines, configuration profiles, compliance policies, and endpoint protection to ensure devices meet organizational standards.

- **Security baselines** — Apply recommended security configurations for Windows endpoints.
- **Configuration profiles** — Define Wi-Fi, VPN, certificates, and device restriction settings.
- **Compliance policies** — Enforce requirements such as BitLocker encryption, minimum OS version, and antivirus status.
- **Endpoint protection** — Configure Microsoft Defender for Endpoint integration.

Deploy Applications

Configure and deploy applications based on organizational needs:

- **Line-of-Business (LoB) apps** — Custom internal applications.
- **Win32 apps** — Packaged desktop applications deployed via .intunewin format.
- **Microsoft Store apps** — Apps distributed through the Microsoft Store for Business.

> **Note:** In most cases, the deployment of applications has already been implemented during the initial infrastructure setup. Ensure all apps and policies are assigned to the group: **"Sec - All Laptop/Desktops"**.

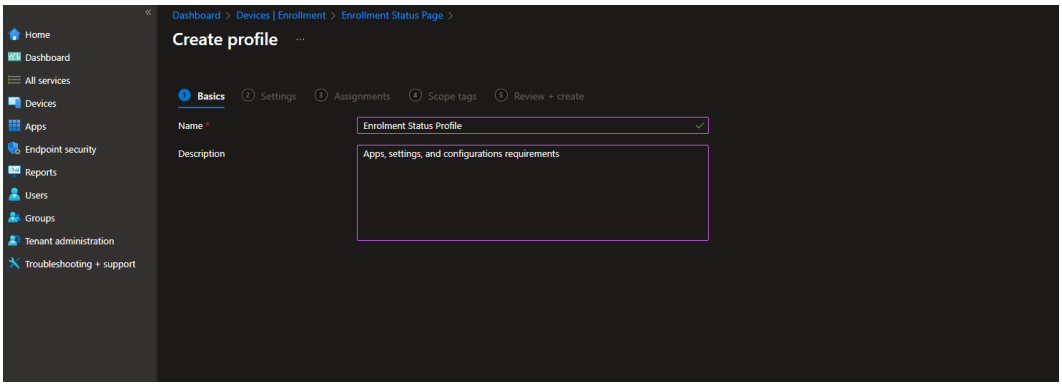
STEP 3 Configure Enrolment Status Page

The Enrolment Status Page (ESP) ensures that all required applications, settings, and configurations are installed and applied before the user can access the desktop during Windows Autopilot deployment. It plays a critical role in ensuring compliance, security, and a smooth user experience.

Navigation Path:

```
Devices > Windows > Windows enrolment > Enrolment Status Page
```

Create a new Enrolment Status Profile. Set the name to **"Enrolment Status Profile"** and add a description such as *"Apps, settings, and configurations requirements"*.



Creating the Enrolment Status Profile

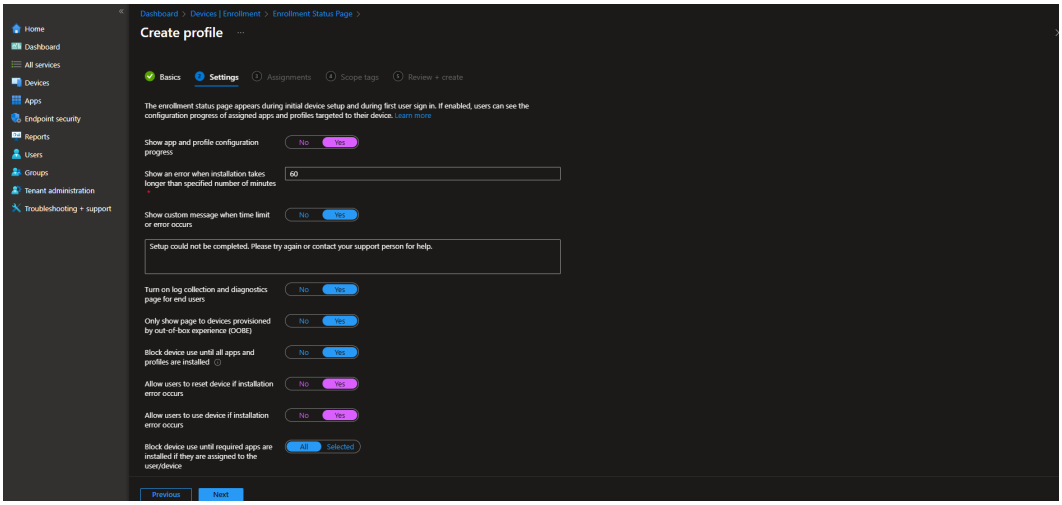
Recommended Settings

Setting	Value	Rationale
Show app and profile config progress	Yes	Users can see deployment progress
Error timeout (minutes)	60	Allows time for large app installs
Show custom message on error	Yes	Provide helpdesk contact info
Log collection and diagnostics	Yes	Aids troubleshooting
Only show for OOBE devices	Yes	Targets Autopilot devices only
Block until all apps installed	Yes	Ensures compliance before access
Allow reset on error	Yes	Enables self-service recovery

Setting	Value	Rationale
Allow use on error	Yes*	Balance UX and compliance
Block until required apps installed	Selected	Critical apps must be present

> **Note:** Enrolment errors can occur due to network glitches. I always enable *"Allow users to reset device if installation error occurs"*. For stricter environments, set *"Allow users to use device if installation error occurs"* to **No** for compliance. The device will sync and complete enrolment in the background.

Assign the profile to the group **"Sec - All Laptop/Desktops"**. Review and create.

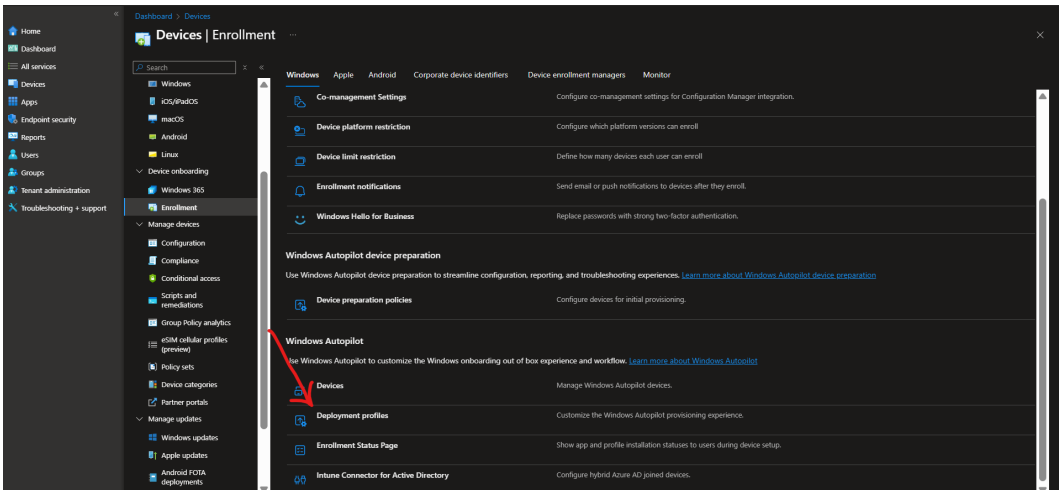


ESP Settings Configuration

STEP 4 Create Deployment Profile

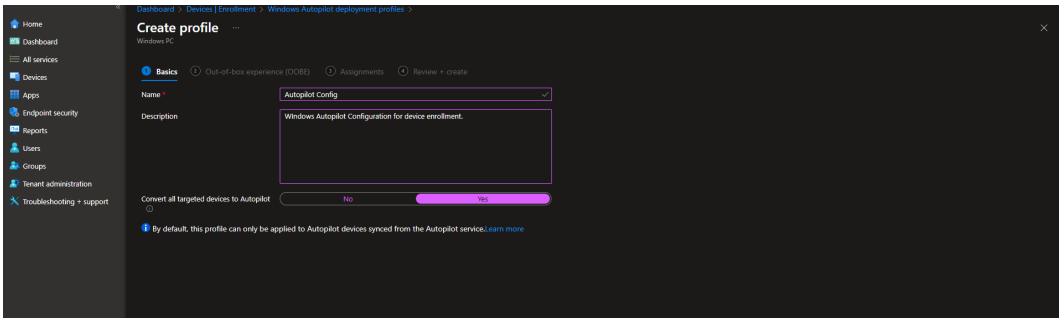
Navigation Path:

`Devices > Windows > Windows enrolment > Deployment Profiles`



Windows Enrolment Overview in Intune

Click **+ Create profile** and select **Windows PC**. Enter a name (e.g., *"Autopilot Config"*) and description (e.g., *"Windows Autopilot Configuration for device enrollment"*).

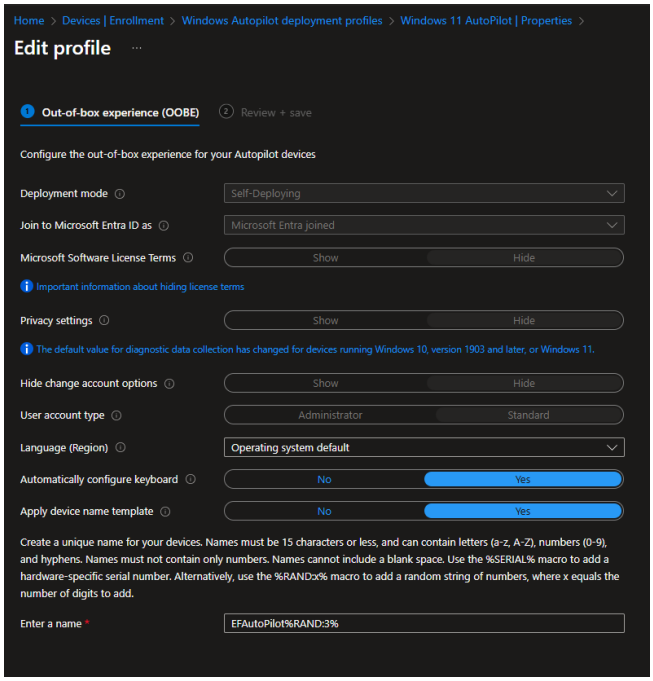


Deployment Profile Basics

Out-of-Box Experience (OOBE) Settings

Setting	Value	Notes
Deployment mode	Self-Deploying	Minimizes user interaction
Join to Entra ID as	Microsoft Entra joined	Auto-handled in self-deploying mode

Setting	Value	Notes
License Terms	Hide	Streamlines the OOBE experience
Privacy settings	Hide	Reduces prompts during setup
Change account options	Hide	Prevents account changes
User account type	Standard	Best practice for security
Language (Region)	OS default	Supports localized preferences
Auto configure keyboard	Yes	Matches keyboard to region
Device name template	Yes	Consistent naming convention



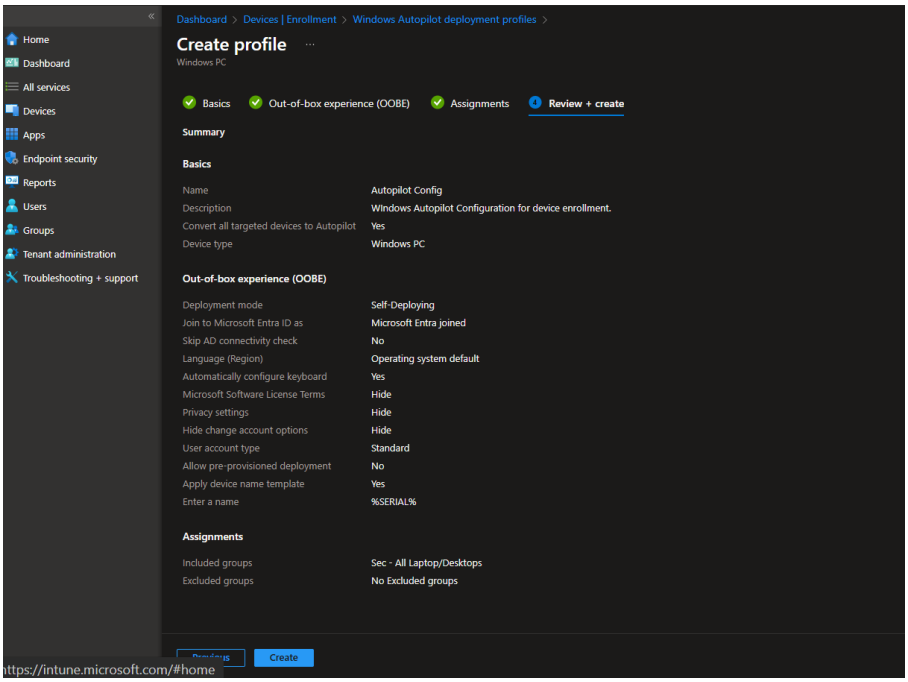
OOBE Configuration Settings

Device Naming Template

Use macros to maintain consistent naming across the organization:

- **%SERIAL%** — Uses the device's hardware serial number.
- **%RAND:x%** — Generates a random string of x digits.

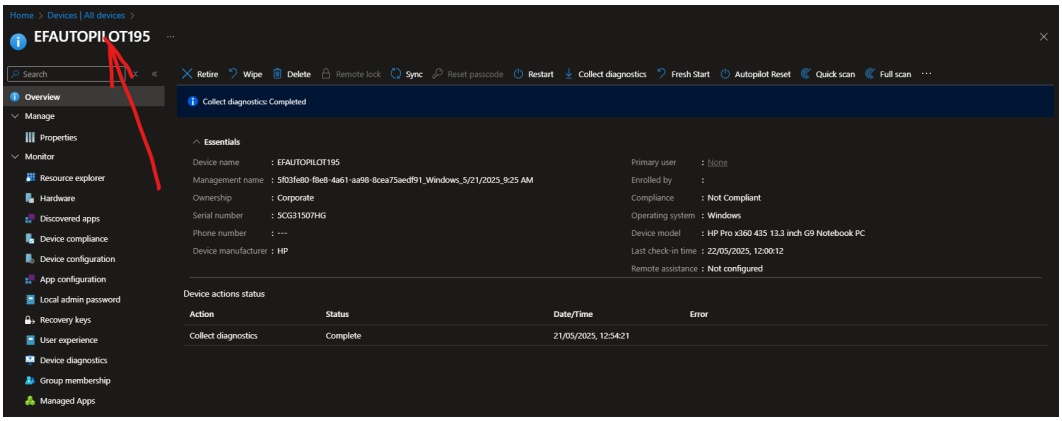
> **Example:** `EFAutoPilot%RAND:3%` produces device names like **EFAUTOPILOT195**.



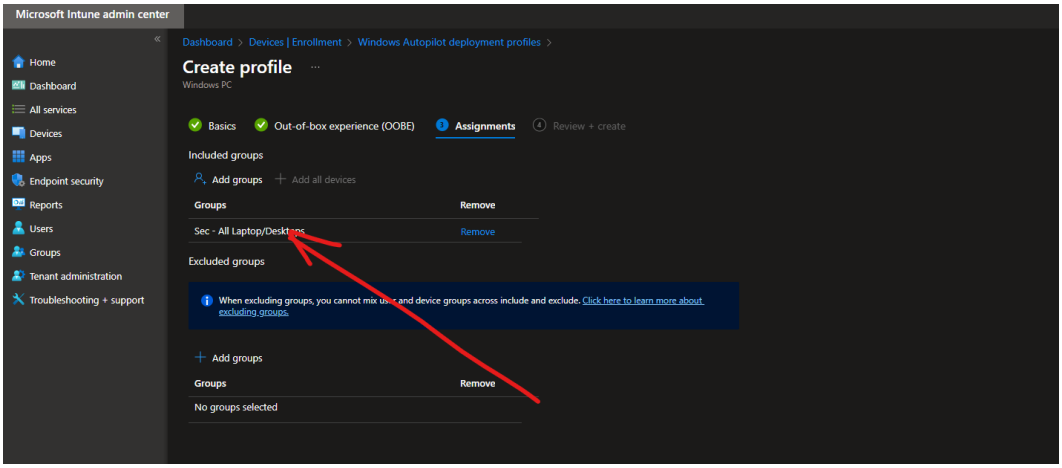
Enrolled Device Example: EFAUTOPILOT195

Assign and Review

Assign the deployment profile to the group **"Sec - All Laptop/Desktops"**.



Profile Assignment to Device Group



Final Review and Create

FINAL STEP Hardware ID Registration

Before enrolling a device via Autopilot, you must upload its **Hardware ID (HWID)**.

- **Bulk purchases:** OEMs can upload the hardware hashes directly to your Autopilot service.
- **Individual devices:** You will need to extract and upload them manually using PowerShell.

> **Reference:** For detailed steps on collecting hardware IDs, see Microsoft's documentation:
> Collecting the device hardware ID (HWID) for Windows Autopilot enrolment

Prepared by:

Olu Adelokiki
Lead EUC Engineer | UserCompute LTD
www.usercompute.com